ENGINEERING MATHEMATICS-II

Department of Science and Humanities

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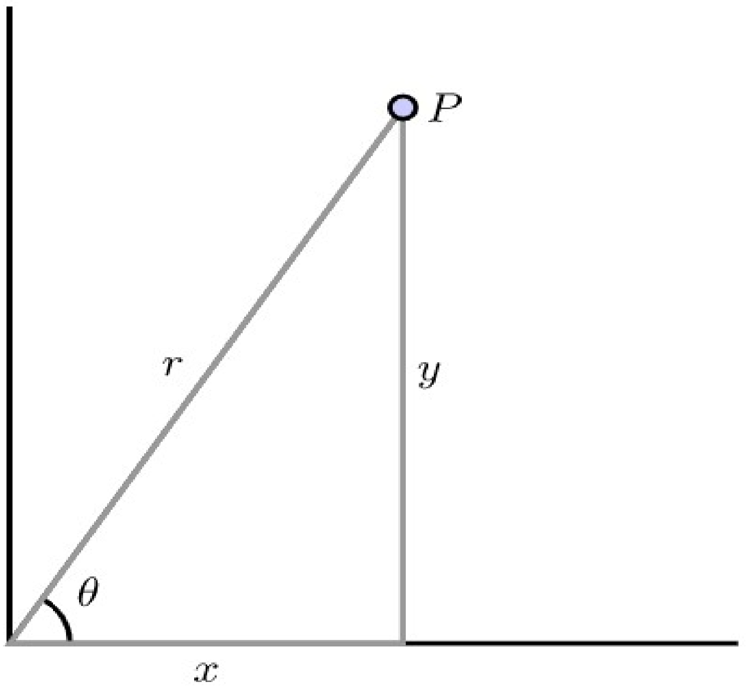
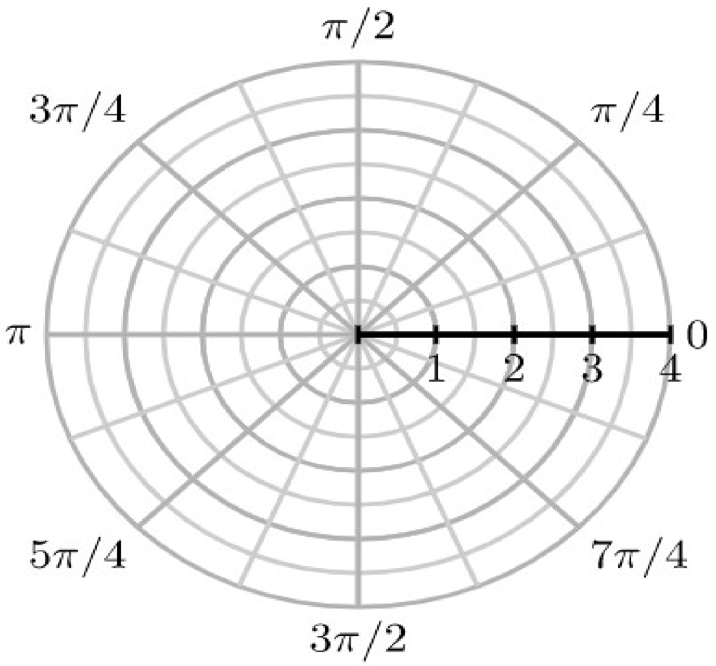
UNIT 1: INTEGRAL CALCULUS

Class 4: problems on change of variables in double integral

POLAR COORDINATES

The rectangular coordinate system is best suited for graphs and regions that are naturally considered over a rectangular grid. The polar coordinate system is an alternative that offers good options for functions and domains that have more circular characteristics. A point P in rectangular coordinates that is described by an ordered pair (x, y), where x is the displacement from P to the y-axis and y is the displacement from P to the x-axis





CARTESIAN TO POLAR COORDINATES

Cartesian to polar coordinates, we have $x = r\cos\theta$ *and* $y = r\sin\theta$.

The Jacobian of x *and* y *with respect to* r *and* θ *is*

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = r$$

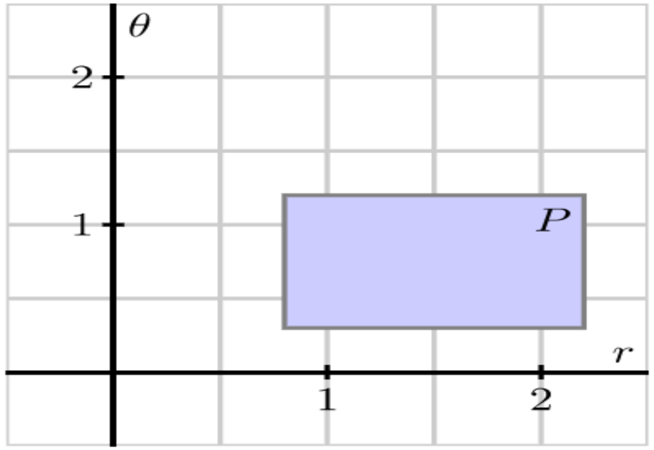
CARTESIAN TO POLAR COORDINATES

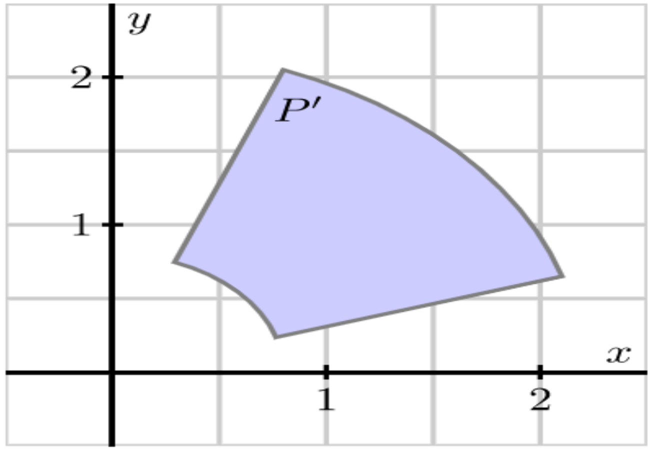
we have

$$\iint_R f(x,y)\,dxdy = \iint_S f(r\cos\theta, r\sin\theta)\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| drd\theta$$

$$= \iint_S f(r\cos\theta, r\sin\theta)\, r\,drd\theta$$

where S is the region in the $r\theta$-plane corresponding to R in the xy-plane.





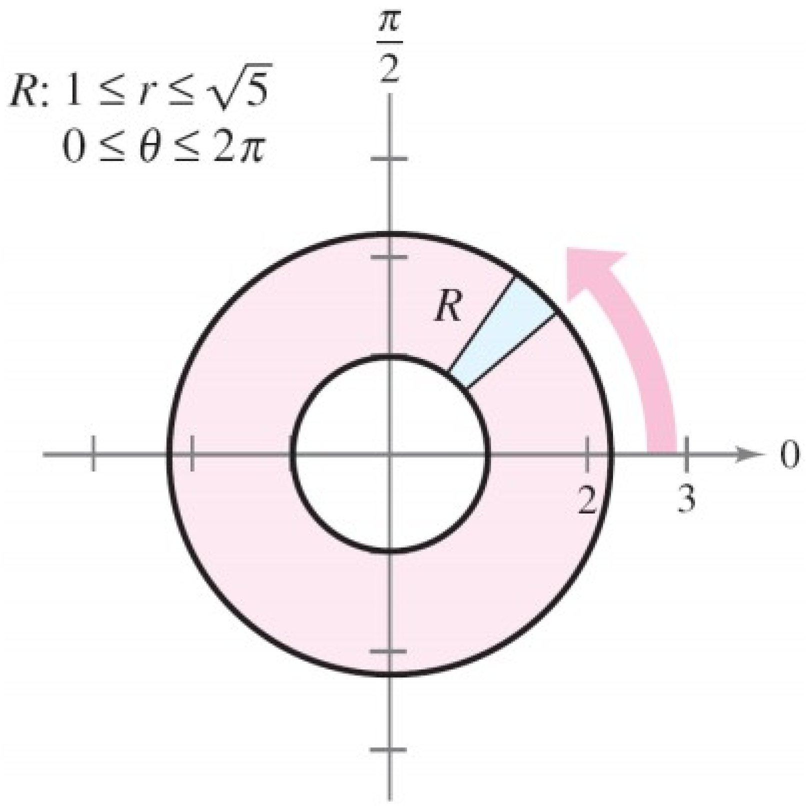
WHEN TO CONVERT FROM CARTESIAN TO POLAR

While there is no firm rule for when polar coordinates can or should be used, they are a natural alternative anytime the domain of integration may be expressed simply in polar form, and/or when the integrand involves expressions such as $\sqrt{x^2 + y^2}$.

Let R be the annular region lying between the two circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 5$. Evaluate the integral $\int_R\int (x^2 + y)\, dA$.

Solution:

The polar boundaries are $1 \le r \le \sqrt{5}$ and $0 \le \theta \le 2\pi$,



PROBLEMS ON POLAR COORDINATES CONTD...

Furthermore, $x^2 = (r\cos\theta)^2$ and $y = r\sin\theta.$

So, you have

$$\int\!\!\int_R (x^2 + y)\, dA = \int_0^{2\pi}\int_1^{\sqrt{5}} (r^2\cos^2\theta + r\sin\theta) r\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_1^{\sqrt{5}} (r^3\cos^2\theta + r^2\sin\theta)\, dr\, d\theta$$

$$= \int_0^{2\pi} \left(\frac{r^4}{4}\cos^2\theta + \frac{r^3}{3}\sin\theta\right)\Big]_1^{\sqrt{5}} d\theta$$

$$= \int_0^{2\pi} \left(6\cos^2\theta + \frac{5\sqrt{5}-1}{3}\sin\theta\right) d\theta$$

$$= \int_0^{2\pi} \left(3 + 3\cos 2\theta + \frac{5\sqrt{5}-1}{3}\sin\theta\right) d\theta$$

$$= \left(3\theta + \frac{3\sin 2\theta}{2} - \frac{5\sqrt{5}-1}{3}\cos\theta\right)\Big]_0^{2\pi}$$

$$= 6\pi.$$

EXAMPLES OF CHANGE OF VARIABLES IN DOUBLE INTEGRAL

Calculate the double integral $\iint\limits_R \sin\sqrt{x^2+y^2}\,dxdy$ by transforming to polar coordinates. The region R is the disk $x^2+y^2 \le \pi^2$.

Solution.

The region R is presented in Figure 8.

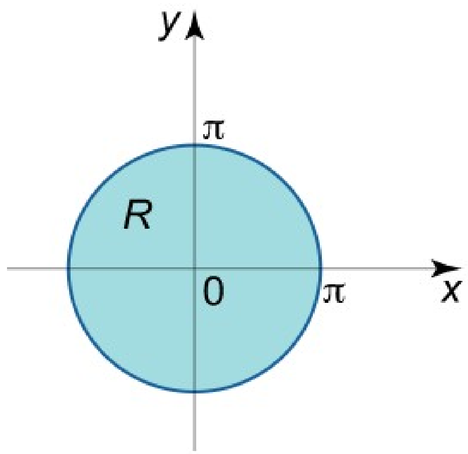


Figure 8.

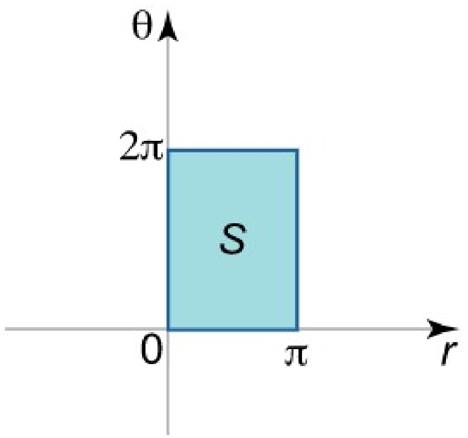


Figure 9.

The image S of the initial region R is defined by the set

$$\{S = (r,\theta) \mid 0 \le r \le \pi,\ 0 \le \theta \le 2\pi\}.$$

EXAMPLE CONTD…

and is shown in Figure 9. The double integral in polar coordinates becomes

$$I = \iint\limits_R \sin\sqrt{x^2+y^2}\,dxdy = \iint\limits_S r\sin r\,dr d\theta = \int\limits_0^{2\pi} d\theta \int\limits_0^{\pi} r\sin r\,dr = 2\pi\int\limits_0^{\pi} r\sin r\,dr.$$

We compute this integral using integration by parts:

$$\int\limits_a^b u dv = (uv)|_a^b - \int\limits_a^b v du.$$

Let $u = r$, $dv = \sin r dr$. Then $du = dr$, $v = \int \sin r dr = -\cos r$. Hence,

$$I = 2\pi\int\limits_0^{\pi} r\sin r\,dr = 2\pi\left[(-r\cos r)|_0^{\pi} - \int\limits_0^{\pi}(-\cos r)\,dr\right]$$

$$= 2\pi\left[(-r\cos r)|_0^{\pi} + \int\limits_0^{\pi}\cos r\,dr\right] = 2\pi\left[(-r\cos r)|_0^{\pi} + (\sin r)|_0^{\pi}\right]$$

$$= 2\pi(\sin r - r\cos r)|_0^{\pi} = 2\pi\left[(\sin\pi - \pi\cos\pi) - (\sin 0 - 0\cdot\cos 0)\right] = 2\pi\cdot\pi = 2\pi^2.$$

EXAMPLE ON POLAR COORDINATES

Let $f(x,y) = e^{x^2+y^2}$ on the disk $D = \{(x,y) : x^2 + y^2 \le 1\}$. evaluate $\iint_D f(x,y)\,dA$.

In rectangular coordinates the double integral $\iint_D f(x,y)\,dA$ can be written as the iterated integral

$$\iint_D f(x,y)\,dA = \int_{x=-1}^{x=1} \int_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}} e^{x^2+y^2}\,dy\,dx.$$

We cannot evaluate this iterated integral, because $e^{x^2+y^2}$ does not have an elementary antiderivative with respect to either x or y. However, since $r^2 = x^2 + y^2$ and the region D is circular, it is natural to wonder whether converting to polar coordinates will allow us to evaluate the new integral. To do so, we replace x with $r\cos(\theta)$, y with $r\sin(\theta)$, and $dy\,dx$ with $r\,dr\,d\theta$ to obtain

$$\iint_D f(x,y)\,dA = \iint_D e^{r^2}\,r\,dr\,d\theta.$$

EXAMPLE CONTD…

The disc D is described in polar coordinates by the constraints $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$. Therefore, it follows that

$$\iint_D e^{r^2}\, r\, dr\, d\theta = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} e^{r^2}\, r\, dr\, d\theta.$$

We can evaluate the resulting iterated polar integral as follows:

$$\begin{aligned}
\int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} e^{r^2}\, r\, dr\, d\theta &= \int_{\theta=0}^{2\pi} \left(\frac{1}{2} e^{r^2} \Big|_{r=0}^{r=1} \right) d\theta \\
&= \frac{1}{2} \int_{\theta=0}^{\theta=2\pi} (e-1)\; d\theta \\
&= \frac{1}{2}(e-1) \int_{\theta=0}^{\theta=2\pi} d\theta \\
&= \frac{1}{2}(e-1)\, [\theta] \Big|_{\theta=0}^{\theta=2\pi} \\
&= \pi(e-1).
\end{aligned}$$

THANK YOU